

Find the Eigen values and Eigen vectors of the matrix

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

1. Characteristic equation $|A - \lambda I| = 0$

$$\begin{aligned} A - \lambda I &= \begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{bmatrix} \\ |A - \lambda I| &= \begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0 \\ &= (-2-\lambda)[(-\lambda)(-\lambda) - (-6)(-2)] - 2[(-\lambda)(-2) - (-6)(-1)] - 3[(-2)(-1) - (-6)(-2)] \\ &= (-2-\lambda)(\lambda^2 - \lambda - 12) - 2(2\lambda - 6) - 3(2 - 12) \\ &= (-2-\lambda)(\lambda^2 - \lambda - 12) - 4\lambda + 12 - 6\lambda + 36 \\ &= (-2-\lambda)(\lambda^2 - \lambda - 12) - 2\lambda + 48 \\ &= (-2-\lambda)(\lambda^2 - \lambda - 12) + 2\lambda + 46 = 0 \end{aligned}$$

2. Eigen vectors

(a) For $\lambda = 5$

Solve $(A - 5I) \vec{v} = \vec{0}$

$$A - 5I = \begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix}$$

Row reduce $\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ 0 & 0 & 0 \end{bmatrix} \sim 0$

From $x - 2y - 3z = 0 \Rightarrow x = 2y + 3z$

Let $y = s, z = t$ (free parameters)

Then $x = 2s + 3t$

Eigen vector for $\lambda = 5$ is $\begin{bmatrix} 2 \\ s \\ 3 \\ t \end{bmatrix} = s \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

Eigen vector for $\lambda = 5$ is any non-zero multiple of $\vec{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ or equivalently $\begin{bmatrix} 12 \\ 6 \\ 0 \\ 0 \end{bmatrix}$

(a) For $\lambda = 3$

Solve $(A - 3I) \vec{v} = \vec{0}$

$$A - 3I = \begin{bmatrix} -5 & 2 & -3 \\ 2 & -2 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

Row reduce $\begin{bmatrix} -5 & 2 & -3 \\ 2 & -2 & -6 \\ 0 & 0 & 0 \end{bmatrix} \sim 0$

From $x - 2y - 3z = 0 \Rightarrow x = 2y + 3z$

Let $y = s, z = t$ (free parameters)

Then $x = 2s + 3t$

Eigen vector for $\lambda = 3$ is $\begin{bmatrix} 2 \\ s \\ 3 \\ t \end{bmatrix} = s \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

Eigen vector for $\lambda = 3$ is any non-zero multiple of $\vec{v}_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

Final Answer:

Eigen values:
 $\lambda = 5$ (mult. 1)
 $\lambda = 3$ (mult. 2)

Eigen vectors:
 For $\lambda = 5$:
 $\text{Span} \left\{ \begin{bmatrix} 12 \\ 6 \\ 0 \\ 0 \end{bmatrix} \right\}$
 For $\lambda = 3$:
 $\text{Span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

1. Determine the rank of matrix

$$A = \begin{bmatrix} 1 & 5 & 6 \\ 2 & 3 & 4 \\ -1 & 2 & 2 \end{bmatrix}$$

Solution:

Use row operations to obtain row echelon form.

$$A = \begin{bmatrix} 1 & 5 & 6 \\ 2 & 3 & 4 \\ -1 & 2 & 2 \end{bmatrix} \xrightarrow{R_2 - R_1, R_3 + R_1} \begin{bmatrix} 1 & 5 & 6 \\ 0 & -7 & -8 \\ 0 & 7 & 8 \end{bmatrix} \xrightarrow{R_3 + R_2} \begin{bmatrix} 1 & 5 & 6 \\ 0 & -7 & -8 \\ 0 & 0 & 0 \end{bmatrix}$$

There are 2 non-zero rows
 $\therefore \text{Rank}(A) = 2$

2. Find the characteristic equation of the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

Solution:

The characteristic equation is given by

$$|A - \lambda I| = 0$$

Compute the determinant

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix}$$

$$= (2-\lambda)[(1-\lambda)(2-\lambda) - 1] = (2-\lambda)[(2-\lambda)^2 - 1] = (2-\lambda)(4 - 4\lambda + \lambda^2 - 1) = (2-\lambda)(\lambda^2 - 4\lambda + 3) = (2-\lambda)(\lambda-1)(\lambda-3)$$

3. Find the centre and radius of the sphere

$$2x^2 + 2y^2 + 2z^2 - 2x + 4y + 2z + 3 = 0$$

Solution:

Divide the equation by 2:

$$x^2 + y^2 + z^2 - x + 2y + z + \frac{3}{2} = 0$$

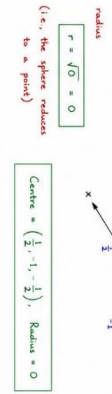
Complete the squares:

$$(x^2 - x) + (y^2 + 2y) + (z^2 + z) = -\frac{3}{2}$$

$$(x - \frac{1}{2})^2 - \frac{1}{4} + (y + 1)^2 - 1 + (z + \frac{1}{2})^2 - \frac{1}{4} = -\frac{3}{2}$$

$$(x - \frac{1}{2})^2 + (y + 1)^2 + (z + \frac{1}{2})^2 = -\frac{3}{2} + \frac{1}{4} + 1 + \frac{1}{4} = 0$$

This is a sphere with
 Centre $C(\frac{1}{2}, -1, -\frac{1}{2})$
 radius $r = \sqrt{0} = 0$
 (i.e. the sphere reduces to a point)



Q.1. Use Green's theorem to evaluate

$$\oint_C [(y - \sin x) dx + \cos x dy]$$

where C is the triangle enclosed by the lines

$$y = 0, \quad x = \frac{\pi}{2}, \quad y = \frac{2x}{\pi}$$

Given,

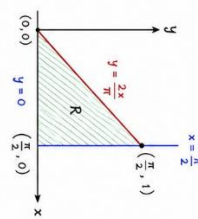
$$P(x, y) = y - \sin x, \quad Q(x, y) = \cos x$$

Green's Theorem:

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$\frac{\partial Q}{\partial x} = -\sin x, \quad \frac{\partial P}{\partial y} = 1$$

$$\therefore \oint_C [(y - \sin x) dx + \cos x dy] = \iint_R (-\sin x - 1) dA = -\iint_R (\sin x + 1) dA \dots \dots \dots$$



Q.2. Evaluate:

(1)

$$I = \int_0^{\pi/2} \int_0^{\cos \theta} re^{-r \cos \theta} dr d\theta$$

Solution:

First integrate w.r.t. θ :

$$I = \int_0^{\pi/2} r \left[\int_0^{\cos \theta} e^{-r \cos \theta} dr \right] d\theta$$

$$\text{Let } t = r \cos \theta \Rightarrow dt = -r \sin \theta d\theta$$

$$\text{When } \theta = 0, \quad t = r; \quad \text{when } \theta = \frac{\pi}{2}, \quad t = 0$$

$$\therefore \int_0^{\pi/2} e^{-r \cos \theta} d\theta = \int_r^0 e^{-t} \left(\frac{-dt}{r \sin \theta} \right) = \frac{1}{r} \int_r^0 e^{-t} dt$$

$$= \frac{1}{r} [-e^{-t}]_r^0 = \frac{1}{r} (1 - e^{-r})$$

$$\text{Hence, } I = \int_0^{\pi/2} r \cdot \frac{1}{r} (1 - e^{-r}) dr = \int_0^{\pi/2} (1 - e^{-r}) dr$$

$$= [r + e^{-r}]_0^{\pi/2} = (\frac{\pi}{2} + 0) - (0 + 1) = \frac{\pi}{2} - 1$$

$$\therefore \int_0^{\pi/2} \int_0^{\cos \theta} re^{-r \cos \theta} dr d\theta = \frac{\pi}{2} - 1 \quad (\text{Divergent})$$

$$\therefore \int_0^{\pi/2} \int_0^{\cos \theta} e^{-r \cos \theta} dr d\theta = \frac{\sqrt{\pi}}{2}$$

$$\text{Let } I = \int_0^{\pi/2} e^{-r \cos \theta} d\theta$$

$$\text{Then } I^2 = \left(\int_0^{\pi/2} e^{-r \cos \theta} d\theta \right) \left(\int_0^{\pi/2} e^{-r \cos \phi} d\phi \right)$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} e^{-(r \cos \theta + r \cos \phi)} d\theta d\phi$$

$$\text{Change to polar coordinates: } x = r \cos \theta, \quad y = r \sin \theta, \quad dx dy = r dr d\theta$$

$$\text{Limits: } r: 0 \text{ to } \infty, \quad \theta: 0 \text{ to } \frac{\pi}{2}$$

$$I^2 = \int_0^{\pi/2} \int_0^{\pi/2} e^{-r \cos \theta} e^{-r \cos \phi} r dr d\theta d\phi$$

$$= \int_0^{\pi/2} \left[-\frac{1}{r} e^{-r \cos \theta} \right]_0^{\pi/2} d\theta = \int_0^{\pi/2} \left[\frac{1}{r} \right]_0^{\pi/2} d\theta = \frac{1}{2} [\theta]_0^{\pi/2} = \frac{\pi}{4}$$

$$\therefore I^2 = \frac{\pi}{4} \Rightarrow I = \frac{\sqrt{\pi}}{2}$$

$$\text{Hence, } \int_0^{\pi/2} e^{-r \cos \theta} d\theta = \frac{\sqrt{\pi}}{2}$$

Evaluate the surface integral

$$\iint_S \vec{F} \cdot \vec{n} dS$$

where

$$\vec{F} = (x^2 + y^2 + z^2)(\vec{i} + \vec{j} + \vec{k})$$

and S is the tetrahedron bounded by

$$x = 0, \quad y = 0, \quad z = 0, \quad x + y + z = 2$$

with outward unit normal.

Using Divergence Theorem:

$$\iint_S \vec{F} \cdot \vec{n} dS = \iiint_V (\nabla \cdot \vec{F}) dV$$

where V is the volume of the tetrahedron.

Compute $\nabla \cdot \vec{F}$:

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \\ &= \frac{\partial}{\partial x} (x^2 + y^2 + z^2) + \frac{\partial}{\partial y} (x^2 + y^2 + z^2) + \frac{\partial}{\partial z} (x^2 + y^2 + z^2) \\ &= 2x + 2y + 2z = 2(x + y + z) \end{aligned}$$

Region V:

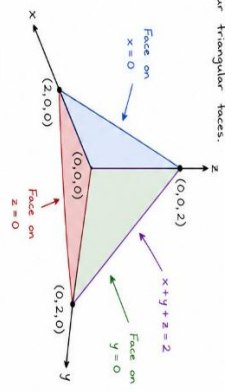
$$V = \{ (x, y, z) : x \geq 0, y \geq 0, z \geq 0, x + y + z \leq 2 \}$$

It is a tetrahedron with intercepts 2 on each axis.

$$\text{Volume of } V = \frac{1}{6} \times 2 \times 2 \times 2 = \frac{4}{3}$$

Solution:

The surface S is the closed tetrahedron in the first octant bounded by the coordinate planes and the plane $x + y + z = 2$. It has four triangular faces.



Now evaluate:

$$\therefore \iint_S \vec{F} \cdot \vec{n} dS = \iiint_V 2(x + y + z) dV = 2 \iiint_V (x + y + z) dV$$

By symmetry of V about the planes,

$$\iiint_V x dV = \iiint_V y dV = \iiint_V z dV$$

Thus,

$$\iiint_V (x + y + z) dV = 3 \iiint_V x dV$$

We know for this tetrahedron,

$$\iiint_V x dV = \frac{1}{4} \times (\text{Volume of } V) \times (x\text{-intercept}) = \frac{1}{4} \times \frac{4}{3} \times 2 = \frac{2}{3}$$

Therefore,

$$\iiint_V (x + y + z) dV = 3 \times \frac{2}{3} = 2$$

$$\therefore \iint_S \vec{F} \cdot \vec{n} dS = 2 \times 2 = 4$$

